The Environment and Directed Technical Change

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November 6, 2025

1. Introduction

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Ouverture

Ouverture

- Aghion's contribution to economic growth theory.
- Growth theory was mathematically formalized in the 1940s and 1950s with the Harrod-Domar and Solow-Swan models: non-optimizing models of long-term economic performance.
- In the 1970s and 1980s, macroeconomics focused more on business cycle fluctuations.
- In the 1990s and 2000s, economic growth slowed down: why?
  1. Schumpeterian growth - Creative Destruction.
  2. Directed technological change.

Ouverture

Figure: Schumpeter

Figure: Aghion

Figure: Acemoglu

Ouverture

1. **Creative Destruction $\rightarrow$ when to innovate:**
   1.1 Creation: An entrepreneur introduces a new technology, obtaining *monopoly* rights.
   1.2 Destruction: This new technology makes old technologies obsolete.
   1.3 Winners and Losers: This process creates new wealth and jobs for those involved in the new technology but displaces the firms and workers tied to the old methods.
2. **Directed technological change $\rightarrow$ where to innovate:**
   2.1 Explains why innovation is "directed" toward specific areas.
   2.2 Technological progress isn't random; it's an endogenous process (meaning it's determined by forces within the economic system) driven by profit motives.
3. **Growth theory and climate change**:
   3.1 Climate change is a threat to economic growth.
   3.2 Green policies can slow down growth.

Introduction

Introduction - General picture

- Growth is determined by technological change.
- Technological change is due to firms' decisions about where to invest resources and researchers: innovation is a random process that depends on these elements.
- Example: dirty and clean innovation.
- Firm decisions about innovation and productions can be socially suboptimal due to externalities.
- Firms invest in polluting technologies rather than clean technologies: the long-term consequences of contamination reduce society's welfare.

Introduction - Questions

1. What are the main determinants of firms' decisions when investing in dirty or clean technologies?
2. Will firms without environmental policies switch to clean technologies?
3. Which policies are sufficient to induce firms to switch to clean innovation?
4. What is the social cost of these policies? What is the social cost of doing nothing?

Introduction

- Two-sector model of directed technological change: one good is produced using dirty inputs (which cause environmental degradation) and the other using clean inputs.
- **Profit-maximizing researchers** drive technological change.
- Technological change depends on the **market size** and **price effects**.
- The relative magnitudes of these effects are determined by:
  1. **Elasticity of substitution** between clean and dirty goods.
  2. **Relative levels of technological development** in the two sectors.
  3. Whether dirty inputs use **exhaustible resources**.

Model economy

Model - Utility function

- Households have preferences of the form

$$\sum_{t=0}^{\infty} \frac{1}{(1+\rho)^t} u(C_t, S_t) \tag{1}$$

- where
  1. C_t is consumption.
  2. S_t is the environmental quality.
  3. $\overline{S}$ is the environmental quality in absence of human pollution.
  4. $\frac{1}{1+\rho}$ is the discount factor.
- u is increasing in C and S, twice continuously differentiable, and concave, and satisfies the Inada conditions:

$$\lim_{C \to 0} \frac{\partial u(C,S)}{\partial C} = \lim_{S \to 0} \frac{\partial u(C,S)}{\partial S} = \lim_{S \to 0} u(C,S) = -\infty \tag{2}$$

- And

$$\frac{\partial u(C, \overline{S})}{\partial S} = 0 \tag{3}$$

Model - Production Function

- The final ouput is a combination of total clean and dirty inputs:

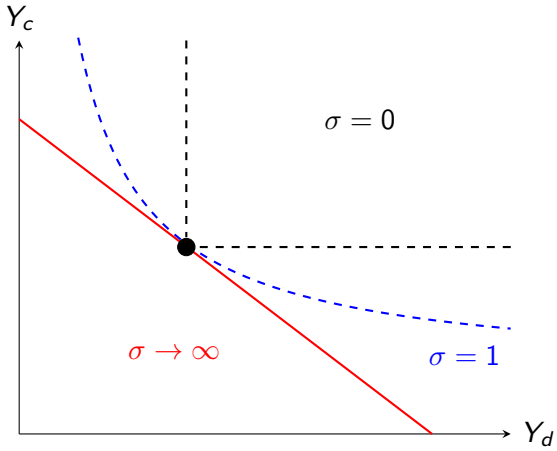
$$Y_t = \left[(Y_t^c)^{\frac{\varepsilon-1}{\varepsilon}} + (Y_t^d)^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}} \tag{4}$$

- Substitutability is given by $\varepsilon \in (0, \infty)$.
- Dirty and clean aggregate inputs $j \in \{d, c\}$ are aggregations of intermediate machines x_{ji} with $i \in [0, 1]$, labor L_j, and the productivity of each machine A_{ji}:

$$Y_t^c = L_{ct}^{1-\alpha} \int_0^1 A_{cit}^{1-\alpha} x_{cit}^{\alpha} di, \quad Y_t^d = R_t^{\alpha_2} L_{dt}^{1-\alpha} \int_0^1 A_{dit}^{1-\alpha_1} x_{dit}^{\alpha_1} di \tag{5}$$

- $\alpha, \alpha_1, \alpha_2 \in (0, 1)$, and $\alpha_1 + \alpha_2 = \alpha$.

Isoquants of Normalized CES Production Functions



Model - Exhaustible resource

- The exhaustible resource stock Q_t satisfies the following recurrence equation:

$$Q_{t+1} = Q_t - R_t \tag{6}$$

- The extraction cost per unit is given by the non-increasing function

$$c(Q_t)$$

- Assume $\alpha_2 = 0$ so the economy has no exhaustible resource (this assumption will be modified later).

Model - Market clearing conditions I

- The labor market-clearing condition is

$$L_t^c + L_t^d \leq 1 \tag{7}$$

- The goods market-clearing condition is given by

$$C_t = Y_t - \psi\left(\int_0^1 x_{cit}\,di + \int_0^1 x_{dit}\,di\right) - c(Q_t)R_t \tag{8}$$

- ψ is the cost of production of one unit of a machine in terms of final output.

Model - Innovation dynamics I (Directed Technical Research)

- Scientist $i \in [0, 1]$ decides whether to do research in sector $j \in \{d, c\}$ and is allocated to one machine x_{jit} with quality A_{jit}.
- The scientist successfully innovate with probability η_j.
- If so, the productivity of the machine A_{jit} increases to $(1 + \gamma)A_{jit}$.
- The scientist earns a one-period patent, becoming an entrepreneur with monopoly rights.
- If not, the monopoly rights go randomly to an entrepreneur drawn from the pool of potential entrepreneurs who uses the old technology.

Model - Innovation dynamics II

- Scientists are normalized to 1 and the market-clearing condition for scientists is

$$s_t^c + s_t^d \leq 1 \tag{9}$$

- Average productivity in each sector is

$$A_{jt} = \int_0^1 A_{jit} di \tag{10}$$

- In sector j for machine i we have:

$$\mathbb{P}(\text{Machine is updated}) = \mathbb{P}(\text{scientist is assigned}) \cdot \mathbb{P}(\text{innovation occurs}) = \eta_j s_{ji}$$

- Average innovation then evolves as

$$A_{jt} = (1 + \gamma \eta_j s_{jt}) A_{jt-1} \tag{11}$$

Model - Environmental dynamics

- Environmental quality evolves as

$$S_{t+1} = -\xi Y_{dt} + (1+\delta)S_t$$

- With

$$-\xi Y_{dt} + (1+\delta)S_t < 0 \Rightarrow S_{t+1} = 0$$
$$-\xi Y_{dt} + (1+\delta)S_t > \overline{S} \Rightarrow S_{t+1} = \overline{S}$$

- ξ is the rate of environmental degradation.
- δ is the rate of environmental regeneration.

Definition (Environmental disaster)

An environmental disaster occurs if $S_t = 0$ for some $t < \infty$.

Model - Big Picture

1. Closed economy with consumers, entrepreneurs, and researchers.
2. Final output is produced with dirty and clean inputs.
3. Each dirty and clean input is produced using dirty and clean intermediate machines owned by entrepreneurs and corresponding dirty and clean productivities.
4. Researchers allocate their research effort based on expected benefits, which stochastically increases the productivity of the corresponding machine.
5. Production of dirty inputs generates pollution, thereby reducing environmental quality.

Environmental Disaster without Exhaustible Resources

Laissez-Faire Equilibrium

- **Which is going to be the outcome if there are no environmental policies?**
- *Laissez-Faire* - a decentralized equilibrium.
- Simple idea: consumers, entrepreneurs (firms) and scientists are **maximizing** and market-clearing conditions are **satisfied**.
- We will **compare this** with the planning problem under policy intervention.

Laissez-Faire Equilibrium

- Formally...

Definition

An equilibrium is a set of sequences of wages (w_t), inputs prices (p_{jt}), machines prices (p_{jit}), demands for machines (x_{jit}), demands for inputs (Y_{jt}), labor demands (L_{jt}) by input producers $j \in \{c, d\}$, research allocations (s_{dt}, s_{et}), and an environment (S_t) such that:

(i) (p_{jit}, x_{jit}) maximizes profits of the producer of machine i in sector j;

(ii) L_{jt} maximizes profits of producers of input j;

(iii) Y_{jt} maximizes the profits of final good producers;

(iv) (s_{dt}, s_{et}) maximizes the expected profit of a researcher at date t;

(v) the wage w_t and the prices p_{jt} clear the labor and input markets respectively; and

(vi) the evolution of S_t is given by (12).

Laissez-Faire Equilibrium - Analysis

- **Which are the main determinants of this outcome?**
- Where will researchers choose to innovate?
- The expected profit Π_{jt} for a scientist engaged in sector j at time t is

$$\Pi_{jt} = \eta_{jt}(1+\gamma)(1-\alpha)p_{jt}^{\frac{1}{(1-\alpha)}} L_{jt} A_{jt-1}$$

- So, the expected relative benefit between the *clean* and *dirty* sectors is

$$\frac{\Pi_{ct}}{\Pi_{dt}} = \frac{\eta_{ct}}{\eta_{dt}} \times \underbrace{\left(\frac{p_{ct}}{p_{dt}}\right)^{\frac{1}{1-\alpha}}}_{\text{price effect}} \times \underbrace{\frac{L_{ct}}{L_{dt}}}_{\text{market size effect}} \times \underbrace{\frac{A_{ct-1}}{A_{dt-1}}}_{\text{direct productivity effect}}$$

- Effects
  1. Price effect: technological change is directed towards the sector with higher selling prices.
  2. Market size effect: technological change is directed towards the sector with more labor.
  3. Productivity effect: technological change is directed towards the more productive sector.

Laissez-Faire equilibrium

- Denote $\varphi = (1-\alpha)(1-\varepsilon)$.
- Relative expected profit can be written as

$$\frac{\Pi_{ct}}{\Pi_{dt}} = \frac{\eta_c}{\eta_d}\left(\frac{1+\gamma\eta_c s_{ct}}{1+\gamma\eta_d s_{dt}}\right)^{-\varphi-1}\left(\frac{A_{ct-1}}{A_{dt-1}}\right)^{-\varphi}$$

Lemma

Under Laissez-Faire, it is an equilibrium for innovation at to occur in the clean when:

$$\eta_c A_{ct-1}^{-\varphi} > \eta_d (1+\gamma\eta_c)^{\varphi+1} A_{dt-1}^{-\varphi},$$

in the dirty sector only when:

$$\eta_d A_{dt-1}^{-\varphi} > \eta_c (1+\gamma\eta_d)^{\varphi+1} A_{ct-1}^{-\varphi},$$

Laissez-Faire equilibrium

Assumption 1

At $t = 0$ productivity is higher in the dirty sector than in the clean sector. $A_{c0} < A_{d0}$

Proposition 1 - Only dirty growth in the Laissez-Faire

Suppose that $\varepsilon > 1$ and Assumption 1 holds. Then there exists a unique *Laissez-Faire* equilibrium where innovation always occur in the dirty sector only, and the long-run growth rate of dirty input production is $\gamma\eta_d$

Proof.

Idea $\rightarrow$ Relative expected profits can be expressed only on terms of past productivity and parameters. The condition $A_{c0} < A_{d0}$ will move the technological progress only to the dirty sector. $\square$

Directed Technical Change

Proposition 2 - Environmental disaster under *Laissez-Faire*

Suppose that $\varepsilon > 1$ and Assumption 1 holds. Then the *Laissez-Faire* equilibrium aways led to an enviromental disaster.

Proof.

This is a consequence of the unbounded growth of dirty output due to continuous growth of dirty technology. □

- Introduction of a governmental subsidize q_t such that

$$\frac{\Pi_{ct}}{\Pi_{dt}} = \frac{(1+q_t)}{1}\frac{\eta_{ct}}{\eta_{dt}} \times \left(\frac{p_{ct}}{p_{dt}}\right)^{\frac{1}{1-\alpha}} \times \frac{L_{ct}}{L_{dt}} \times \frac{A_{ct-1}}{A_{dt-1}}$$

- Force more profitability of the green sector $\rightarrow$ Increase A_c

Directed Technical Change

Proposition 3 - Environmental disaster and input substitutions

When the two inputs are strong substitutes and $\overline{S}$ is sufficiently high, a temporary subsidy to clean research will prevent an environmental disaster. In contrast, when the two inputs are weak substitutes, a temporary subsidy to clean research cannot prevent an environmental disaster.

Optimal Environmental Policy without Exhaustible Resources

Socially Optimal Allocation

- **Necessary policies to arrive to social optimal**
- To achieve for social optimal allocation is necessary to correct by 2 externalities:
  1. Enviromental externalities due to dirty input producers
  2. The fact that researchers does not internalize the effect of their research in the future.
- Achieved by lump-sum taxes and transfers (subsidies)

Definition - Socially optimal Allocation

The socially optimal allocation is a dynamic path of final good production Y_t, consumption C_t, input productions Y_{jt}, machine productions x_{jit}, labor allocations L_{jt}, scientist allocations s_{jt}, environmental quality S_t, and qualities of machines A_{jit} that maximizes the intertemporal utility of the representative consumer, (1), subject to (4), (5), (7), (8), (9), (11), and (12) (with $R_t \equiv 0$ and $\alpha_2 = 0$). The following proposition is one of our main results.

Socially Optimal Allocation

Proposition 5 - Social optimal allocation and environmental policies

The socially optimal allocation can be implemented using a tax on dirty input (*carbon tax*), a subsidy to clean innovation, and a subsidy for the use of all machines.

- The optimal tax is given by

$$\tau_t = \frac{\omega_{t+1}\xi}{\lambda_t \hat{p}_{dt}},$$

- ω is the costate variable associated with the enviromental quality S_t, λ_t is the costate variable associated with the consumption good C_t, $\hat{p}_{dt}$ if the shadow price of input j (in terms of final ouput)..
- The tax is such that the marginal cost of producing dirty input by one unit must be equal to the resulting marginal benefit in terms of higher enviromental quality in all the next periods.

Optimal allocation of scientists

- Scientists will be allocated towards the sector with higher **social gain** from innovation.
- Research is directed towards clean sector whenever

$$\frac{\eta_c(1+\gamma\eta_c s_{ct})^{-1}\sum_{v\geq t}\frac{\partial u(C_v,S_v)}{\partial C}\frac{1}{(1+\rho)^v}\hat{p}_{cv}^{1/(1-\alpha)}L_{cv}A_{cv}}{\eta_d(1+\gamma\eta_d s_{dt})^{-1}\sum_{v\geq t}\frac{\partial u(C_v,S_v)}{\partial C}\frac{1}{(1+\rho)^v}\hat{p}_{dv}^{1/(1-\alpha)}L_{dv}A_{dv}} > 1$$

- In the decentralized case the decision to do research or not in the clear sector was depending mainly on the blue term.
- For the decentralized case such term represent the individual value of innovation

Structure of Optimal Environmental Regulation

Proposition 6 - Social optimal allocation and environmental policies

Suppose that $\varepsilon > 1$ and the discount rate ρ is suffifciently small. Then all innovation switches to the clean sector in finite time, the economy grows asymptocally at rate $\gamma\eta_c$ and the optimal subsidy on profits in the clean sector, q_t, is temporary. Moreover, if $\varepsilon > \frac{1}{(1-\alpha)}$ then the optimal carbon tax, τ_t is temporary.

- Result: Taxes and subsidies are enough to modify the direction of innovation.
- Policy intervention is not permanent: If $\varepsilon > 1$, there exists a time $t \geq 0$ such that $A_{ct} > A_{dt}$, meaning that all research will eventually be directed toward the clean sector.
- Effectiveness of the policies depends on the substitutability between the two inputs.

Conclusions

Conclusions - non discounted parts

- The case where $\varepsilon < 1$
  1. In this case not policies are useful to mantain economic growth and green transition. The only solution to avoind an enviromental disaster is de-growth.
- Exhaustible resources for the production of dirty inputs
  1. In this scenatio the reduction of the resources increases its price, making less attratcie to produce wiht it and making firms switch to clean inputs.
  2. Is not a real scenario (oil, gas and coal reserves are simple too big)
- Modeling robustness
  1. Pollution affects productivity
  2. Alternative technologies
  3. Alternative laws of motion of environmental stock
- Cost of delay.

Conclusions

- Decentralized markets lead to environmental disaster due to externalities in innovation and dirty production.
- A socially optimal allocation can be achieved with temporary environmental policies: carbon taxes to reduce polluting production and subsidies to incentivize clean research.
- However, the effectiveness of these policies depends on a sufficiently high degree of substitutability between clean and dirty inputs.
- If substitutability is too low, an environmental disaster may be unavoidable.

Discussions

Discussions

All theory depends on assumptions which are not quite true.That is what makes it theory. The art of successful theorizing is to make the inevitable simplifying assumptions in such a way that the final results are not very sensitive. A "crucial" assumption is one on which the conclusions do depend sensitively, and it is important that crucial assumptions be reasonably realistic. When the results of a theory seem to flow specifically from a special crucial assumption, then if the assumption is dubious, the results are suspect

Robert Solow, 1958, *A contribution to the theory of economic growth.*

Discussions - Closed vs. Open Economy

$\rightarrow$ Model's current assumption: closed economy.

$\rightarrow$ Now, assume an open economy where domestic policies cannot affect foreign firms.

- Assume N firms are generating a dirty input.
- $m \in [0, n)$ domestic firms pay taxes, and $m' \in [n, N]$ foreign firms do not.
- The marginal cost for foreign firms (m') will be relatively lower than the marginal cost of domestic firms (m), which affects prices $\rightarrow$ The carbon tax has no effect.
- With a sufficiently high ε (elasticity?) between domestic and foreign firms, the local (m) firms are displaced from the market, and only the m' polluting firms remain.
- Example: imports from developing countries.
- Question: How many m' firms are needed to cause displacement? How quickly does this displacement occur?

Discussions - The ε-Problem

$\rightarrow$ Model assumption: one good.
$\rightarrow$ Reality: $\bar{\varepsilon} = \int_0^1 \varepsilon_i di$

- When discussing clean and dirty goods, there are many examples where ε is very low over non-trivial intervals. The meat industry is a good example.
- We can express this as:

$$Y_t = \int_0^1 \left[(Y_{it}^c)^{\frac{\varepsilon_i - 1}{\varepsilon_i}} + (Y_{it}^d)^{\frac{\varepsilon_i - 1}{\varepsilon_i}} \right]^{\frac{\varepsilon_i}{\varepsilon_i - 1}} di$$

- For some $j \in [0, 1]$, ε_j is very low.

Discussions — The ε Problem

- This means that at least some of the dirty input $f(Y_{tj}^d, \varepsilon_j)$ will still have to be used.
- In the best-case scenario, where all firms have completed the green transition except the one for which elasticity is very low, the environmental situation evolves according to:

$$S_{t+1} = -\xi f(Y_{jt}, \varepsilon_j) + (1 + \delta) S_t$$

- The environmental situation continues to worsen even after the green transition (depending on the relative strength of the two terms in the difference equation).
- Consumer preferences must shift so that good j is no longer demanded.

Discussion – Main Problem

- Two main points raised by the paper:
  1. Different initial levels of production inputs between the dirty and clean sectors.
  2. Competition and entry barriers for clean energy.

Discussion – Main Problem

- The paper focuses on an important obstacle to inducing the Green Transition: the stock dimension of dirty physical and knowledge capital.
- This is not the first case in history—it is the essence of creative destruction.
- However, looking at historical examples, at least one of the following conditions must be satisfied:
  1. A shift in preferences.
  2. A disruptive technology.

Discussion – Institutional Environment

- **Corporate environment**
  1. Polluting companies have more power $\rightarrow$ they can cannibalize clean (new) companies.
  2. How can this be avoided without harming market mechanisms?
- **Political environment**
  1. The Green Transition is incentivized by environmental policies that have short- to medium-term costs (as shown in the paper).
  2. Why should a policymaker (i.e., a politician) be incentivized to adopt measures that are unpopular among the majority of voters?
  3. Mean-field model of voter/politician behavior.

Equilibrium profits

1. Profix maximization of the input producer in sector j at time t.
2. Find the iso-elastic inverse demand curve for each machine.
3. Solve monopolistic machine owner problem.
4. Price is a constant markup over marginal cost.
5. We can find the equilibrium demand for the machine.

Equilibrium profits and scientists allocation

1. Write the expected profits on function of productivity.
2. Study the expression as function of $1 + \varphi$

1. Find λ_t, the shadow value of one unit of final good consumption at time t (and also the shadow value of one unit more of consumption).
2. ω is the Lagrange mutplier of the environmental constraint dynamics.
3. Shadow value of enviromental quality at timr t is equal to the marginal utility that it generates in this period plus the shadow value $(1+\delta)$ units of enviromental quality in the next period.
4. This equation can be solved recursively.
5. Define the lagrande multplier for the input producitons goods
6. Define the ration $\frac{\lambda_j}{\lambda} = \hat{p}_{jt}$
7. FOCs with respect clean and firty input, for the dirty input one finds that the FOCs has the term $\frac{\omega_{t+1}\xi}{\lambda} \Rightarrow \tau_t = \frac{\omega_{t+1}\xi}{\lambda_t \hat{p}_{dt}}$